NATIONAL UNIVERSITY OF SINGAPORE

IS2108 – Full-stack Software Engineering for AI Solutions I

(Semester 1: AY2026/27)

MOCK PRACTICAL EXAM

Paper 1

Time Allowed: 1 Hour 30 Minutes

INSTRUCTIONS TO STUDENTS

1. This assessment paper contains **FIVE** (5) tasks and comprises **SIX** (6) printed pages, including this page.

2. **You are required to attempt all tasks. If your final deliverable cannot be run, method marks may be awarded subjected to satisfactory evidence in the written code and a maximum cap of 50% of the marks allocated to each of the affected task or user story.**

3. **You are allowed to use a Generative AI tool in this paper, but the use is strictly limited to GitHub Copilot. You may activate GitHub Education using your NUS email address.**

4. This is an **OPEN BOOK** assessment. You may only refer to approved offline materials.

5. **You are NOT allowed to use the Internet to refer to any online resources.**

6. At the end of the assessment, return this assessment paper in its entirety. Do **NOT** retain or redistribute the assessment paper.

7. This assessment is conducted with your own laptop, i.e., Bring Your Own Device (BYOD). It is your responsibility to ensure that your laptop has been setup according to the IS2108 hardware and software requirements.

8. Place your Django project folder and the Python virtual environment folder "venv" inside a common parent folder named as "mockpe". Do **NOT** use git in this assessment. At the end of the assessment, compress only the **Django project folder** into a single ZIP archive file. Name the ZIP file as "studentmgtsys.zip". Upload this ZIP file to the Canvas Assignment: Assessments > Mock Practical Exam.

9. **Do NOT submit the Python virtual environment folder "venv" or ".git" folder.**

10. You are required to arrive at your designated venue 20 minutes before the start of the assessment from 6:40 pm onwards to set up your laptop.

This portion is for examiner's use only

Task	Maximum Possible Marks	Marks Obtained	Remark
A	2		
B	3		
C	1		
D	2		
E	6		
Total	**14**		

Task A – Create Django Project (2 marks)

1. Open the parent folder “mockpe” in Visual Studio Code (VS Code).
2. Create a Python virtual environment with the folder name “venv”.
3. Install the following Python packages:
   - Django
4. Export the list of Python packages along with their exact version numbers using the following command. Ensure that “requirements.txt” is in the parent folder “mockpe”.

   ```
   python -m pip freeze > requirements.txt
   ```

5. Create a new Django project for **Student Management System** with the name “studentmgtsys”.
6. Create a new app with the name “adminmodule” in the Django project.
7. Create another new app with the name “studentmodule” in the Django project.
8. The project must follow the standard folder structure as shown below. At the end of the assessment, you should only submit the “studentmgtsys” folder.

   ```
   mockpe
   |
   |--- studentmgtsys
   |    |
   |    |--- adminmodule
   |    |--- studentmgtsys
   |    |--- studentmodule
   |
   |--- venv
   ```

Task B – Create Django Models (3 marks)

1. The UML class diagram in Figure 1 depicts the logical data model of the Student Management System. Create the required Django model classes in the “adminmodule” app to implement the logical data model.

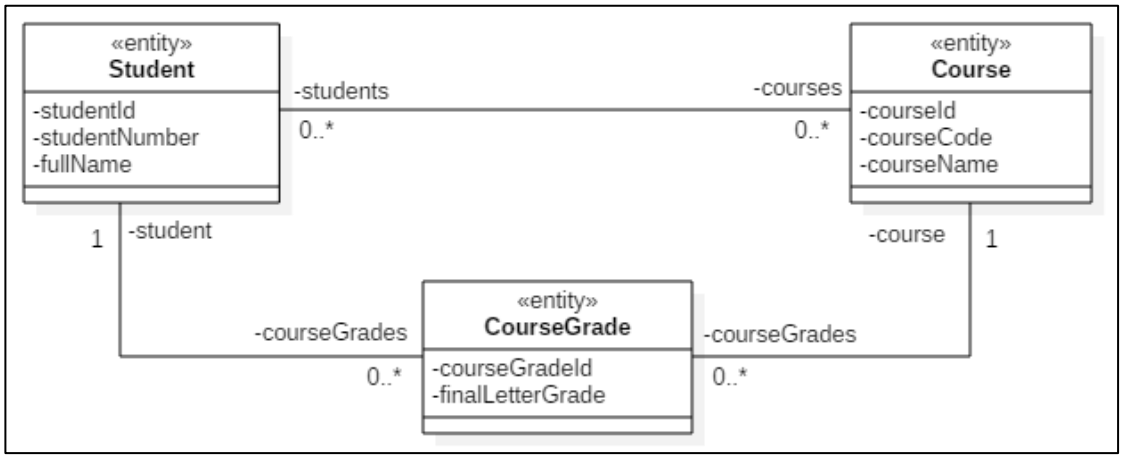


Figure 1 – UML class diagram depicting the logical data model for the Student Management System.

2. Infer the most appropriate field type for each field of a model. Apply the strictest possible integrity constraints in the model classes.

3. You are required to take into consideration the following points:

   a) Primary key for `Student` is `studentId`.
   b) Primary key for `Course` is `courseId`.
   c) Primary key for `CourseGrade` is `courseGradeId`.
   d) All primary key values are numerical whole number and should be auto-generated.
   e) `Student.studentNumber` is required, must be unique and can have a maximum of 9 characters.
   f) `Student.fullName` is required and can have a maximum of 64 characters.
   g) `Course.courseCode` is required, must be unique and can have a maximum of 12 characters.
   h) `Course.courseName` is required and can have a maximum of 128 characters.
   i) `CourseGrade.FinalLetterGrade` is required and can have a maximum of 2 characters.

4. Create the migration files and apply the migrations to the database. Ensure that the database tables are created successfully.

Task C – Test Data Loading (1 marks)

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1. In the "adminmodule" app, create a suitable mechanism to load the following test data for the `Course` model after the web application has been run in the development server.

Course Code	Course Name
IS2108	Full-stack Software Engineering for AI Solutions I
IS2109	AI and Machine Learning Techniques I
IS3108	Full-stack Software Engineering for AI Solutions II
IS3109	AI and Machine Learning Techniques II
IS4108	AI Solutioning Capstone Project
IS4151	AIoT Solutions and Development
IS4400	Human-AI Interaction
IS4401	Generative AI and Business Applications
IS4402	Machine Learning with Unstructured Data

Task D – Web Application Structure, Aesthetics and Navigation (2 marks)

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1. Use a suitable technique to structure all the templates in the web application into four sections – header, left navigation menu, right content, and footer. The schematic layout diagram is shown in Figure 2 below.



Figure 2 – Schematic layout diagram of templates.

2. Create a suitable primary navigation menu according to the sitemap shown in Figure 3 on the next page. The navigation menu should include a "Home" link back to the "index.html" of the respective app.

3. You should also take into consideration the user stories provided in Task E.

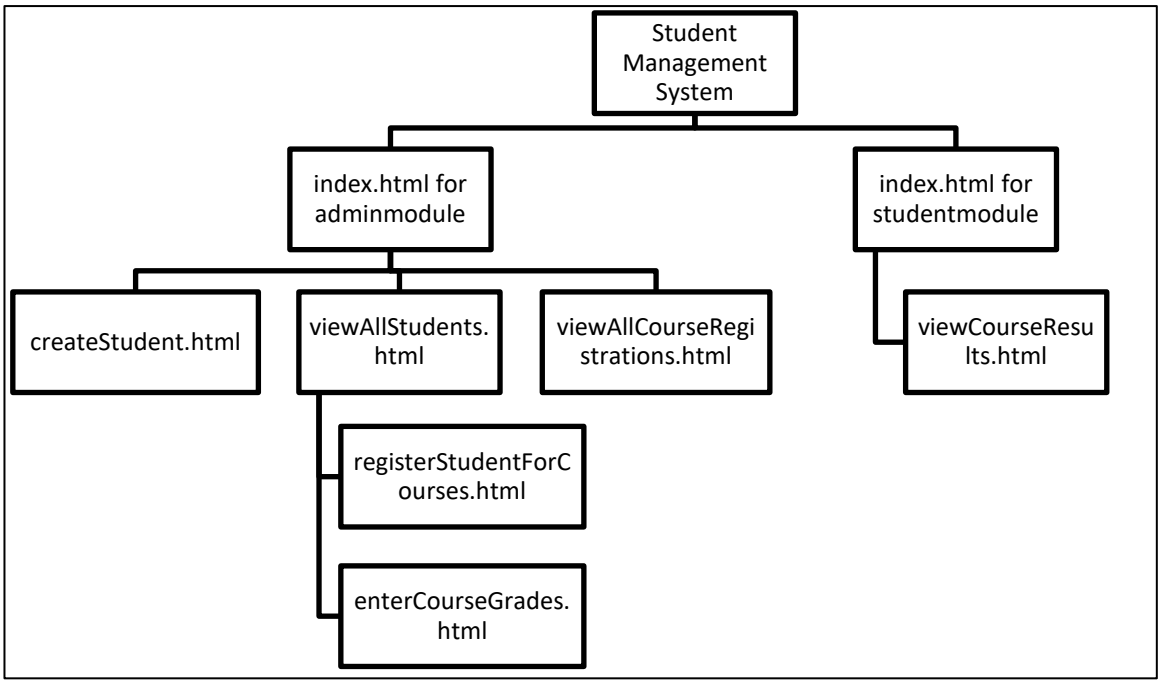


Figure 3 – Sitemap of the Student Management System.

Task E – Implement System Functionalities to Realise Use Stories (6 marks)

1. Perform all development tasks using only Python and Django features to implement system functionalities to realise all the **SIX (6)** user stories depicted in Table 1 below.

Story ID/Title	Story Description	Additional Context
1/ Create Student (1 mark)	As an admin, I would like to create a new student record so his/her course registration information and results can be managed.	• Input student number and full name • Create a new student record. • Print out the student ID of the new student after successful creation by displaying a suitable info message. • If student number already exists, display an error message.
2/ View All Students (1 mark)	As an admin, I would like to view a list of all students so that I can select a particular student to manage his/her course registration information and results.	• Retrieve and display a list of all student records in the system using a table. • For each student record, print out the student ID, student number and full name. • The table should contain an additional "Action" column with two links per student record: ○ Register Student for Courses ○ Enter Course Grades

Story ID/Title	Story Description	Additional Context
3/Register Student for Courses (1 mark)	As an admin, I would like to register a student for one or more courses so s/he can read the courses and graduate.	• Print out the student ID, student number and full name of the selected student. • Display a list of all available courses in a table. • Design and implement a suitable web user interface for admin to select one or more courses to be registered to the student. • Save the course registration information for the student and display a suitable info message.
4/Enter Course Grades (1 mark)	As an admin, I would like to enter the final letter grade of each course that a student has read so his/her results can be used to determine graduation status.	• Print out the student ID, student number and full name of the selected student. • Display a list of all courses registered to the student in a table. • Design and implement a suitable web user interface for admin to input the final letter grade of each course. • Possible letter grades are “A+”, “A”, “A-“, “B+”, “B” , “C+”, “C” , “D+”, “D” , “E” and “F”. • Save the course grades for the student and display a suitable info message.
5/View All Course Registrations (1 mark)	As an admin, I would like to view a list of all course registrations so I can have an overview of all students’ academic career.	• Retrieve and display a list of all course registration records in the system using a table. • For each course registration record, print out student number, student full name, course code and course name. • Group the rows by student, i.e., the registered courses for the same student should be displayed together or across adjacent rows.
6/View Course Results (1 mark)	As a student, I would like to view the final letter grade of each course that I have read so I can monitor my own graduation status.	• Display a list of all courses registered to the student in a table. • For each course registration record, print out course code, course name and the student’s final letter grade for that course

Table 1 – User stories for Student Management System.

2. For all user stories, input data validation should be provided.

- End of Paper 1 -